

U.S. Units

TABLE 15-15 Hollow Tubing: Steel and Aluminum, Square and Rectangular, Smaller Sizes: 0.375 in to 3.00 in Depth

Outside dimensions				Inside dimensions		1Section properties											
Ref.	Short side		Long side	Wall thickness	Short side		Long side		Area	2Weight/ft				Axis X-X		Axis Y-Y	
	a (in)	b (in)		t (in)	c (in)	d (in)	A (in²)	Steel (lb/ft)		Aluminum (lb/ft)	I _x (in⁴)	S _x (in³)	r _x (in)	I _y (in⁴)	S _y (in³)	r _y (in)	
a	0.375	0.375		0.049	0.277	0.277	0.064	0.217	0.077	0.00116	0.00617	0.13458	0.00116	0.00617	0.135		
b	0.500	0.500		0.065	0.370	0.370	0.113	0.384	0.136	0.00365	0.0146	0.180	0.00365	0.0146	0.180		
c	0.500	1.000		0.065	0.370	0.870	0.178	0.605	0.214	0.0214	0.0427	0.346	0.00674	0.0270	0.195		
d	0.500	1.500		0.065	0.370	1.370	0.243	0.826	0.292	0.0613	0.0818	0.502	0.0098	0.0394	0.201		
e	0.500	1.500		0.120	0.260	1.260	0.422	1.434	0.507	0.0973	0.130	0.480	0.0138	0.0551	0.181		
f	0.625	0.625		0.065	0.495	0.495	0.146	0.494	0.175	0.00771	0.0247	0.230	0.00771	0.0247	0.230		
g	0.750	0.750		0.065	0.620	0.620	0.178	0.605	0.214	0.0141	0.0375	0.281	0.0141	0.0375	0.281		
h	0.750	1.500		0.065	0.620	1.370	0.276	0.936	0.331	0.0781	0.104	0.532	0.0255	0.0681	0.304		
i	0.875	0.875		0.049	0.777	0.777	0.162	0.550	0.194	0.0185	0.042	0.338	0.0185	0.0422	0.338		
j	1.000	1.000		0.065	0.870	0.870	0.243	0.826	0.292	0.0356	0.071	0.383	0.0356	0.0712	0.383		
k	1.000	1.000		0.120	0.760	0.760	0.422	1.434	0.507	0.0555	0.111	0.363	0.0555	0.111	0.363		
l	1.000	1.500		0.065	0.870	1.370	0.308	1.046	0.370	0.0948	0.126	0.555	0.0498	0.100	0.402		
m	1.000	1.500		0.120	0.760	1.260	0.542	1.842	0.651	0.155	0.206	0.534	0.0789	0.158	0.381		
n	1.000	2.000		0.065	0.870	1.870	0.373	1.267	0.448	0.193	0.193	0.718	0.0640	0.128	0.414		
o	1.000	3.000		0.065	0.870	2.870	0.503	1.709	0.604	0.536	0.357	1.032	0.0925	0.185	0.429		
p	1.250	1.250		0.065	1.120	1.120	0.308	1.046	0.370	0.0723	0.116	0.485	0.0723	0.116	0.485		
q	1.375	1.375		0.058	1.259	1.259	0.306	1.038	0.367	0.0885	0.129	0.538	0.0885	0.129	0.538		
r	1.500	1.500		0.065	1.370	1.370	0.373	1.267	0.448	0.128	0.171	0.586	0.128	0.171	0.586		
s	1.500	1.500		0.120	1.260	1.260	0.662	2.250	0.795	0.212	0.282	0.566	0.212	0.282	0.566		
t	1.500	2.000		0.065	1.370	1.870	0.438	1.488	0.526	0.253	0.253	0.761	0.162	0.216	0.608		
u	1.500	2.000		0.120	1.260	1.760	0.782	2.657	0.939	0.428	0.428	0.739	0.269	0.359	0.586		
v	1.500	3.000		0.120	1.260	2.760	1.022	3.472	1.227	1.167	0.778	1.069	0.384	0.512	0.613		

Notes: I_y = moment of inertia; S = section modulus; r = radius of gyration, used in column analysis; See sketch for X-X and Y-Y axes.

Values are for perfectly square corners. Some vendors apply radii to inside and/or outside corners.

Wall thickness gages: 0.049 in = 18 gage; 0.058 in = 17 gage; 0.065 in = 16 gage; 0.120 in = 11 gage.

Other gages and sizes are available from many manufacturers and vendors. See Appendix A15-14 for larger structural tubing.

²Using density of steel = 0.283 lb_m/in³, density of aluminum = 0.100 lb_m/in³. (May range from 0.095 lb_m/in³ to 0.102 lb_m/in³.)

Sources for data for additional sizes: Reliance Steel & Aluminum Co./Earl M. Jorgensen Co., multiple locations, Central Steel & Wire Co., multiple locations, OnlineMetals.com, Seattle, WA, Bull Moose Tube Co., Metals Depot, Winchester, KY.

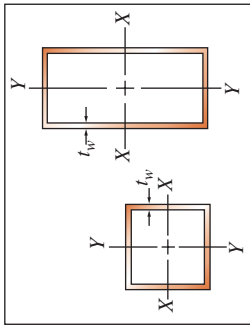


TABLE 15-16 Hollow Tubing: Steel and Aluminum, Square and Rectangular: 20 mm to 300 mm Depth

Ref.	Outside dimensions			Wall			Inside dimensions			Area		² Weight/ft		¹ Section properties					
	Short side		Long side	thickness			Short side		Long side	A (mm ²)	Steel (N/m)	Aluminum (N/m)	I _x (in ⁴)	Axis X-X		Axis Y-Y			
	a (mm)	b (mm)		t (mm)			c (mm)	d (mm)						S _x (mm ³)	I _x (in ⁴)	r _x (mm)	I _y (mm ⁴)	S _y (mm ³)	r _y (mm)
a	10	20		2			6	16		104	7.84	2.82	4.619E+03	462	6.66	1.379E+03	276	3.64	
b	20	20		2			16	16		144	10.8	3.91	7.872E+03	787	7.39	7.872E+03	787	7.39	
c	20	30		3			14	24		264	19.9	7.17	2.887E+04	1925	10.5	1.451E+04	1451	7.41	
d	30	30		3			24	24		324	24.4	8.80	3.985E+04	2657	11.1	3.985E+04	2657	11.1	
e	20	40		3			14	34		324	24.4	8.80	6.081E+04	3041	13.7	1.889E+04	1889	7.64	
f	40	40		3			34	34		444	33.5	12.1	1.020E+05	5099	15.2	1.020E+05	5099	15.2	
g	30	50		3			24	44		444	33.5	12.1	1.421E+05	5685	17.9	6.181E+04	4121	11.8	
h	50	50		3			44	44		564	42.5	15.3	2.085E+05	8340	19.2	2.085E+05	8340	19.2	
i	40	80		4			32	72		896	67.5	24.3	7.113E+05	1.778E+04	28.2	2.301E+05	1.150E+04	16.0	
j	80	80		3			74	74		924	69.6	25.1	9.145E+05	2.286E+04	31.5	9.145E+05	2.286E+04	31.5	
k	50	100		4			42	92		1136	85.6	30.8	1.441E+06	2.883E+04	35.6	4.737E+05	1.895E+04	20.4	
l	100	100		4			92	92		1536	116	41.7	2.363E+06	4.727E+04	39.2	2.363E+06	4.727E+04	39.2	
m	50	150		4			42	142		1536	116	41.7	4.041E+06	5.388E+04	51.3	6.858E+05	2.743E+04	21.1	
n	150	150		5			140	140		2900	218	78.7	1.017E+07	1.357E+05	59.2	1.017E+07	1.357E+05	59.2	
o	200	200		4			192	192		3136	236	85.1	2.009E+07	2.009E+05	80.0	2.009E+07	2.009E+05	80.0	
p	100	200		4			92	192		2336	176	63.4	1.240E+07	1.240E+05	72.9	4.208E+06	8.415E+04	42.4	
q	50	200		4			42	192		1936	146	52.6	8.561E+06	8.561E+04	66.5	8.979E+05	3.592E+04	21.5	
r	250	250		8			234	234		7744	583	210.3	7.567E+07	6.054E+05	98.9	7.567E+07	6.054E+05	98.9	
s	300	300		8			284	284		9344	704	253.7	1.329E+08	8.859E+05	119.3	1.329E+08	8.859E+05	119.3	
t	300	300		12.5			275	275		14375	1083	390.3	1.984E+08	1.323E+06	117.5	1.984E+08	1.323E+06	117.5	

Notes: I_y = moment of inertia; S = section modulus; r = radius of gyration, used in column analysis; See sketch for X-X and Y-Y axes.

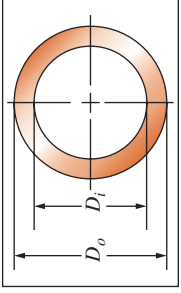
Values are for perfectly square corners. Some vendors apply radii to inside and/or outside corners.

Numerous additional sizes and wall thicknesses are available from many manufacturers and vendors.

²Using density of steel = 7680 kg/m³; density of aluminum = 2768 kg/m³. (May range from 2635 kg/m³ to 2829 kg/m³.)

Sources for data for additional sizes: Parker Steel Company—Metric Sized Metals, Toledo, Ohio.

Bull Moose Tube Co.

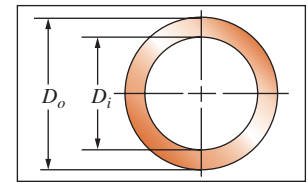


U.S. Units

TABLE 15–17 Pipe: American National Standard Schedule 40 and AISC Standard: 1/8 in to 18 in Sizes

Ref.	¹ Nominal size (in)	Outside diameter (in)	Inside diameter (in)	Wall thickness <i>t_w</i> (in)	Area <i>A</i> (in ²)	Weight/ft Steel (lb/ft)	Section properties				
							<i>I</i> (in ⁴)	<i>S</i> (in ³)	<i>r</i> (in)	Torsional constants	
										<i>J</i> (in ⁴)	<i>Z_p</i> (in ³)
Schedule 40 pipe											
a	1/8 in	0.405	0.269	0.068	0.072	0.245	1.06E-03	5.25E-03	0.122	2.13E-03	1.05E-02
b	1/4 in	0.540	0.364	0.088	0.125	0.425	3.31E-03	1.23E-02	0.163	6.62E-03	2.45E-02
c	3/8 in	0.675	0.493	0.091	0.167	0.568	7.29E-03	2.16E-02	0.209	1.46E-02	4.32E-02
d	PIPE 1/2 STD	0.840	0.622	0.109	0.250	0.851	1.71E-02	4.07E-02	0.261	3.42E-02	8.14E-02
e	PIPE 3/4 STD	1.050	0.824	0.113	0.333	1.131	3.70E-02	7.05E-02	0.334	7.41E-02	0.1411
f	PIPE 1 STD	1.315	1.049	0.133	0.494	1.679	8.73E-02	0.1328	0.421	0.1747	0.2657
g	PIPE 1-1/4 STD	1.660	1.380	0.140	0.669	2.273	0.1947	0.2346	0.540	0.3894	0.4692
h	PIPE 1-1/2 STD	1.900	1.610	0.145	0.799	2.718	0.3099	0.3262	0.623	0.6198	0.6524
i	PIPE 2 STD	2.375	2.067	0.154	1.075	3.653	0.6657	0.5606	0.787	1.331	1.121
j	PIPE 2-1/2 STD	2.876	2.469	0.203	1.704	5.793	1.530	1.064	0.947	3.059	2.128
k	PIPE 3 STD	3.500	3.068	0.216	2.228	7.576	3.017	1.724	1.164	6.034	3.448
l	PIPE 3-1/2 STD	4.000	3.548	0.226	2.680	9.109	4.788	2.394	1.337	9.575	4.788
m	PIPE 4 STD	4.500	4.026	0.237	3.714	10.790	7.233	3.214	1.510	14.47	6.429
n	PIPE 5 STD	5.563	5.047	0.258	4.300	14.618	15.16	5.451	1.878	30.32	10.90
o	PIPE 6 STD	6.625	6.065	0.280	5.581	18.974	28.14	8.496	2.245	56.28	16.99
p	PIPE 8 STD	8.625	7.981	0.322	8.399	28.554	72.49	16.81	2.938	145.0	33.62
q	PIPE 10 STD	10.750	10.020	0.365	11.908	40.483	160.7	29.90	3.674	321.5	59.81
r	12 in	12.750	11.938	0.406	15.745	53.525	300.2	47.09	4.367	600.4	94.18
s	16 in	16.000	15.000	0.500	24.347	82.771	731.9	91.49	5.483	1464	183.0
t	18 in	18.000	16.876	0.562	30.788	104.667	1171	130.2	6.168	2343	260.3

Notes: ¹All values shown are for standard schedule 40 steep pipe.
Rows d–q conform to AISC standards for dimensions of standard weight pipe, Rows a–c and r–t do not.
Many other sizes of round hollow structural sections (HSS) are available. See AISC Manual.
Sources for data for additional sizes: American Piping Products, Chesterfield, MO.
Davidson Group, Specialty Pipe & Tube, Brooklyn, NY.



U.S. Units

TABLE 15-18 Mechanical Tubing: Steel and Aluminum: 0.5 in to 5.0 in Outside Diameter

Ref.	Nominal size		Outside diameter D_o (in)	Inside diameter D_i (in)	Wall thickness t_w (in)	Area A (in ²)	² Weight/ft		¹ Section properties				
									Bending properties		Torsion properties		
	OD (in)	Wall gauge					Steel (lb/ft)	Aluminum (lb/ft)	I (in ⁴)	S (in ³)	r (in)	J (in ⁴)	Z_p (in ³)
a	1/2	17	0.500	0.384	0.058	0.081	0.274	0.097	0.00200	0.00800	0.158	0.00400	0.0160
b	1/2	14	0.500	0.334	0.083	0.109	0.370	0.130	0.00246	0.00983	0.150	0.00491	0.0197
c	1	16	1.000	0.870	0.065	0.191	0.649	0.229	0.0210	0.0419	0.331	0.0419	0.0839
d	1	10	1.000	0.732	0.134	0.365	1.239	0.437	0.0350	0.0700	0.310	0.0700	0.140
e	1 1/2	16	1.500	1.370	0.065	0.293	0.996	0.352	0.0756	0.101	0.508	0.151	0.202
f	1 1/2	10	1.500	1.232	0.134	0.575	1.955	0.690	0.135	0.181	0.485	0.271	0.361
g	2	16	2.000	1.870	0.065	0.395	1.343	0.474	0.185	0.185	0.685	0.370	0.370
h	2	10	2.000	1.732	0.134	0.786	2.671	0.943	0.344	0.344	0.661	0.687	0.687
i	2 1/2	10	2.500	2.232	0.134	0.996	3.386	1.195	0.699	0.559	0.838	1.398	1.119
j	2 1/2	5	2.500	2.060	0.220	1.576	5.357	1.891	1.034	0.827	0.810	2.067	1.654
k	3	10	3.000	2.732	0.134	1.207	4.102	1.448	1.241	0.828	1.014	2.483	1.655
l	3	5	3.000	2.560	0.220	1.921	6.532	2.306	1.868	1.245	0.986	3.736	2.490
m	3 1/2	10	3.500	3.232	0.134	1.417	4.817	1.700	2.010	1.149	1.191	4.020	2.297
n	3 1/2	5	3.500	3.060	0.220	2.267	7.707	2.720	3.062	1.750	1.162	6.125	3.500
o	4	5	4.000	3.560	0.220	2.613	8.882	3.135	4.682	2.341	1.339	9.364	4.682
p	4	3/8 in	4.000	3.250	0.375	4.271	14.518	5.125	7.090	3.545	1.288	14.180	7.090
q	4 1/2	5	4.500	4.060	0.220	2.958	10.056	3.550	6.791	3.018	1.515	13.583	6.037
r	4 1/2	3/8 in	4.500	3.750	0.375	4.860	16.521	5.832	10.422	4.632	1.464	20.843	9.264
s	5	5	5.000	4.560	0.220	3.304	11.231	3.964	9.456	3.782	1.692	18.911	7.564
t	5	3/8 in	5.000	4.250	0.375	5.449	18.523	6.538	14.665	5.866	1.641	29.329	11.732

Notes: ¹ I = moment of inertia; S = section modulus; r = radius of gyration, used in column analysis; J = polar moment of inertia; Z_p = polar section modulus.

²Using density of steel = 0.283 lb_m/in³; density of aluminum = 0.100 lb_m/in³. (May range from 0.095 lb_m/in³ to 0.102 lb_m/in³.)

Sources for data for additional sizes: Reliance Steel & Aluminum Co./Earl M. Jorgensen Co.

Central Steel & Wire Co.

Webco Industries, Inc.

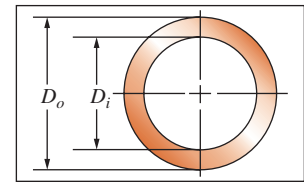
Bull Moose Tube Co.

Paramount Extrusions, Co., Paramount, CA.

Metals Depot, Winchester, KY.

Wheatland Tube Company, Sharon, PA (smaller sizes).

Davidson Group, Specialty Pipe & Tube, Brooklyn, NY (larger sizes and heavier wall).



SI Units

TABLE 15-19 Mechanical Tubing: Steel and Aluminum: 10 mm to 150 mm Outside Diameters

Ref.	Outside diameter D_o (mm)	Inside diameter D_i (mm)	Wall thickness t (mm)	Area A (mm ²)	² Weight/m		¹ Section properties				
					Steel (N/m)	Aluminum (N/m)	Bending properties		r (mm)	Torsion properties	
							I (mm ⁴)	S (mm ³)		J (mm ⁴)	Z_p (mm ³)
a	10	8	1.0	28.27	2.130	0.768	289.8	57.96	3.20	579.6	115.9
b	10	6	2.0	50.27	3.787	1.365	427.3	85.45	2.92	854.5	170.9
c	20	17	1.5	87.18	6.568	2.367	3754	375.4	6.56	7508.3	750.8
d	20	15	2.5	137.4	10.36	3.732	5369	536.9	6.25	1.074E+04	1074
e	30	26	2.0	175.9	13.25	4.777	1.733E+04	1155	9.92	3.466E+04	2311
f	30	22	4.0	326.7	24.62	8.872	2.826E+04	1884	9.30	5.652E+04	3768
g	45	40	2.5	333.8	25.15	9.064	7.563E+04	3361	15.05	1.513E+05	6722
h	45	37	4.0	515.2	38.82	13.990	1.093E+05	4857	14.56	2.186E+05	9715
i	60	52	4.0	703.7	53.02	19.109	2.773E+05	9242	19.85	5.545E+05	1.848E+04
j	60	48	6.0	1017.9	76.69	27.639	3.756E+05	1.252E+04	19.21	7.512E+05	2.504E+04
k	75	70	2.5	569.4	42.90	15.462	3.746E+05	9.988E+03	25.65	7.491E+05	1.998E+04
l	75	65	5.0	1100	82.84	29.857	6.769E+05	1.805E+04	24.81	1.354E+06	3.610E+04
m	90	84	3.0	820	61.78	22.265	7.767E+05	1.726E+04	30.78	1.553E+06	3.452E+04
n	90	80	5.0	1335	100.6	36.255	1.210E+06	2.689E+04	30.10	2.420E+06	5.378E+04
o	110	104	3.0	1008	76.0	27.383	1.444E+06	2.626E+04	37.85	2.889E+06	5.252E+04
p	110	100	5.0	1649	124.3	44.786	2.278E+06	4.142E+04	37.17	4.556E+06	8.284E+04
q	130	124	3.0	1197	90.2	32.502	2.415E+06	3.715E+04	44.91	4.829E+06	7.429E+04
r	130	120	5.0	1963	147.9	53.317	3.841E+06	5.909E+04	44.23	7.682E+06	1.182E+05
s	150	144	3.0	1385	104.4	37.620	3.744E+06	4.992E+04	51.98	7.488E+06	9.983E+04
t	150	140	5.0	2278	171.6	61.847	5.993E+06	7.991E+04	51.30	1.199E+07	1.598E+05

Notes: ¹ I = moment of inertia; S = section modulus; r = radius of gyration, used in column analysis; J = polar moment of inertia; Z_p = polar section modulus.

²Using density of steel = 7680 kg/m³; density of aluminum = 2768 kg/m³. (May range from 2635 kg/m³ to 2829 kg/m³.)

Source for data for additional sizes: Parker Steel Company—Metric Sized Metals, Toledo, Ohio.

APPENDIX 16 Conversion Factors

Mass Standard SI unit: Kilogram (kg). Equivalent unit: $\text{N} \cdot \text{s}^2/\text{m}$.

$\frac{14.59 \text{ kg}}{\text{slug}}$	$\frac{32.174 \text{ lb}_m}{\text{slug}}$	$\frac{2.205 \text{ lb}_m}{\text{kg}}$	$\frac{453.6 \text{ grams}}{\text{lb}_m}$	$\frac{2000 \text{ lb}_m}{\text{ton}_m}$	$\frac{1000 \text{ kg}}{\text{metric ton}_m}$
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Force Standard SI unit: Newton (N). Equivalent unit: $\text{kg} \cdot \text{m}/\text{s}^2$.

$\frac{4.448 \text{ N}}{\text{lb}_f}$	$\frac{10^5 \text{ dynes}}{\text{N}}$	$\frac{4.448 \times 10^5 \text{ dynes}}{\text{lb}_f}$	$\frac{224.8 \text{ lb}_f}{\text{kN}}$	$\frac{1000 \text{ lb}}{\text{K}}$
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Length

$\frac{3.281 \text{ ft}}{\text{m}}$	$\frac{39.37 \text{ in}}{\text{m}}$	$\frac{12 \text{ in}}{\text{ft}}$	$\frac{25.4 \text{ mm}}{\text{in}}$	$\frac{1.609 \text{ km}}{\text{mi}}$	$\frac{5280 \text{ ft}}{\text{mi}}$
-------------------------------------	-------------------------------------	-----------------------------------	-------------------------------------	--------------------------------------	-------------------------------------

Area

$\frac{144 \text{ in}^2}{\text{ft}^2}$	$\frac{10.76 \text{ ft}^2}{\text{m}^2}$	$\frac{645.2 \text{ mm}^2}{\text{in}^2}$	$\frac{10^6 \text{ mm}^2}{\text{m}^2}$	$\frac{43\,560 \text{ ft}^2}{\text{acre}}$	$\frac{10^4 \text{ m}^2}{\text{hectare}}$
--	---	--	--	--	---

Volume

$\frac{1728 \text{ in}^3}{\text{ft}^3}$	$\frac{231 \text{ in}^3}{\text{gal}}$	$\frac{7.48 \text{ gal}}{\text{ft}^3}$	$\frac{264 \text{ gal}}{\text{m}^3}$	$\frac{3.785 \text{ L}}{\text{gal}}$	$\frac{35.3 \text{ ft}^3}{\text{m}^3}$
---	---------------------------------------	--	--------------------------------------	--------------------------------------	--

Pressure, Stress, or Loading Standard SI unit: Pascal (Pa). Equivalent units: N/m^2 or $\text{kg}/\text{m} \cdot \text{s}^2$.

$\frac{144 \text{ lb}/\text{ft}^2}{\text{lb}/\text{in}^2}$	$\frac{47.88 \text{ Pa}}{\text{lb}/\text{ft}^2}$	$\frac{6895 \text{ Pa}}{\text{lb}/\text{in}^2}$	$\frac{1 \text{ Pa}}{\text{N}/\text{m}^2}$	$\frac{6.895 \text{ MPa}}{\text{ksi}}$
--	--	---	--	--

Energy Standard SI unit: Joule (J). Equivalent units: $\text{N} \cdot \text{m}$ or $\text{kg} \cdot \text{m}^2/\text{s}^2$.

$\frac{1.356 \text{ J}}{\text{lb} \cdot \text{ft}}$	$\frac{1.0 \text{ J}}{\text{N} \cdot \text{m}}$	$\frac{8.85 \text{ lb} \cdot \text{in}}{\text{J}}$	$\frac{1.055 \text{ kJ}}{\text{Btu}}$	$\frac{3.600 \text{ kJ}}{\text{W} \cdot \text{h}}$	$\frac{778 \text{ ft} \cdot \text{lb}}{\text{Btu}}$
---	---	--	---------------------------------------	--	---

Power Standard SI unit: Watt (W). Equivalent unit: J/s or $\text{N} \cdot \text{m}/\text{s}$.

$\frac{745.7 \text{ W}}{\text{hp}}$	$\frac{1.0 \text{ W}}{\text{N} \cdot \text{m}/\text{s}}$	$\frac{550 \text{ lb} \cdot \text{ft}/\text{s}}{\text{hp}}$	$\frac{1.356 \text{ W}}{\text{lb} \cdot \text{ft}/\text{s}}$	$\frac{3.412 \text{ Btu}/\text{h}}{\text{W}}$	$\frac{1.341 \text{ hp}}{\text{kW}}$
-------------------------------------	--	---	--	---	--------------------------------------

Density (Mass/Unit Volume)

$\frac{515.4 \text{ kg}/\text{m}^3}{\text{slug}/\text{ft}^3}$	$\frac{1000 \text{ kg}/\text{m}^3}{\text{gram}/\text{cm}^3}$	$\frac{32.17 \text{ lb}_m/\text{ft}^3}{\text{slug}/\text{ft}^3}$	$\frac{16.018 \text{ kg}/\text{m}^3}{\text{lb}_m/\text{ft}^3}$
---	--	--	--

Specific Weight (Weight/Unit Volume)

$\frac{157.1 \text{ N}/\text{m}^3}{\text{lb}_f/\text{ft}^3}$	$\frac{1728 \text{ lb}/\text{ft}^3}{\text{lb}/\text{in}^3}$
--	---

Bending Moment or Torque

$\frac{8.851 \text{ lb} \cdot \text{in}}{\text{N} \cdot \text{m}}$	$\frac{1.356 \text{ N} \cdot \text{m}}{\text{lb} \cdot \text{ft}}$
--	--

Section Modulus

$\frac{1.639 \times 10^4 \text{ mm}^3}{\text{in}^3}$	$\frac{10^9 \text{ mm}^3}{\text{m}^3}$
--	--

Moment of Inertia or Second Moment of an Area

$\frac{4.162 \times 10^5 \text{ mm}^4}{\text{in}^4}$	$\frac{10^{12} \text{ mm}^4}{\text{m}^4}$
--	---

General approach to applying conversion factors: Arrange the conversion factor from this table such that, when multiplied by the given quantity, the original units cancel out, leaving the desired units. See examples below.

Example 1. Convert a stress of 36 ksi to MPa.

$$\sigma = 36 \text{ ksi} \times \frac{6.895 \text{ MPa}}{\text{ksi}} = 248 \text{ MPa}.$$

Example 2. Convert a stress of 1272 MPa to ksi.

$$\sigma = 1272 \text{ MPa} \times \frac{1.0 \text{ ksi}}{6.895 \text{ MPa}} = 184 \text{ ksi}.$$

APPENDIX 17 Hardness Conversion Table

Brinell	Rockwell		Vickers	Steel: tensile strength, (1000 psi approx.)	Brinell	Rockwell		Vickers	Steel: tensile strength, (1000 psi approx.)
No. ¹	B	C	No. ²		No. ¹	B	C	No. ²	
(745)		65.3			262	(103.0)	26.6	277	127
(712)		63.3			255	(102.0)	25.4	268	123
(682)		61.7			248	(101.0)	24.2	261	120
(653)		60.0			241	100.0	22.8	252	116
(627)		58.7			235	99.0	21.7	246	114
601		57.3	639		229	98.2	20.5	241	111
578		56.0	614		223	97.3	(18.8)	235	108
555		54.7	590	298	217	96.4	(17.5)	228	105
534		53.5	569	288	212	95.5	(16.0)	222	102
514		52.1	546	274	207	94.6	(15.2)	217	100
495		51.6	527	269	201	93.8	(13.8)	211	98
477		50.3	508	258	197	92.8	(12.7)	207	95
461		48.8	490	244	192	91.9	(11.5)	202	93
444		47.2	472	231	187	90.7	(10.0)	196	90
429		45.7	455	219	183	90.0	(9.0)	184	89
415		44.5	440	212	179	89.0	(8.0)	188	87
401		43.1	424	202	174	87.8	(6.4)	183	85
388		41.8	410	193	170	86.8	(5.4)	179	83
375		40.4	396	184	167	86.0	(4.4)	175	81
363		39.1	383	177	163	85.0	(3.3)	171	79
352	(110.0)	37.9	371	171	156	82.9	(0.9)	164	76
341	(109.0)	36.6	360	164	149	80.8			73
331	(108.5)	35.5	349	159	143	78.7			71
321	(108.0)	34.3	338	154	137	76.4			67
311	(107.5)	33.1	327	149	131	74.0			65
302	(107.0)	32.1	319	146	126	72.0			63
293	(106.0)	30.9	309	141	121	69.8			60
285	(105.5)	29.9	301	138	116	67.6			58
277	(104.5)	28.8	292	134	111	65.7			56
269	(104.0)	27.6	284	130					

Notes: This is a condensation of Table 2, *Report J417b, SAE 1971 Handbook*. Values in () are beyond normal range and are presented for information only.

¹Values above 500 are for tungsten carbide ball; below 500 are for standard ball.

²Vickers: Diamond pyramid Hardness Number; 50 kg load.

APPENDIX 18 Stress Concentration Factors

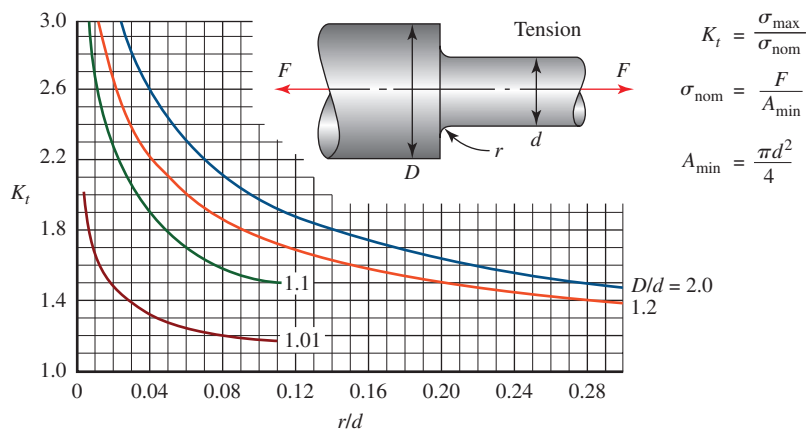


FIGURE A18-1 Stepped round bar in tension

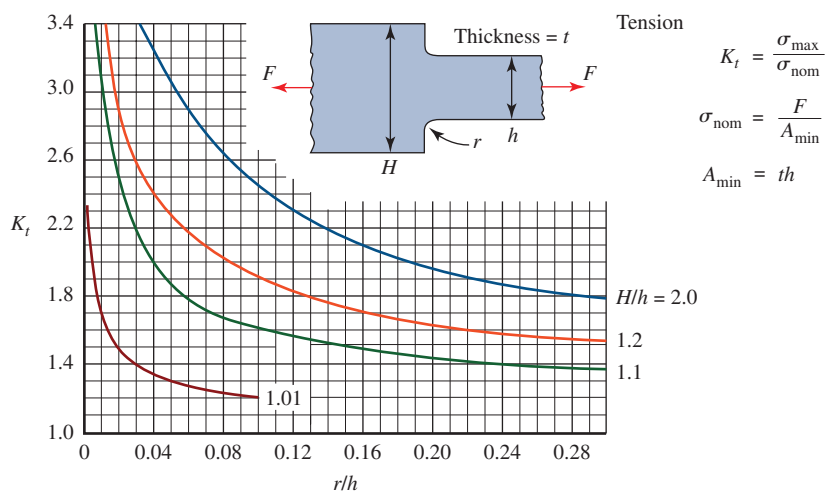


FIGURE A18-2 Stepped flat plate in tension

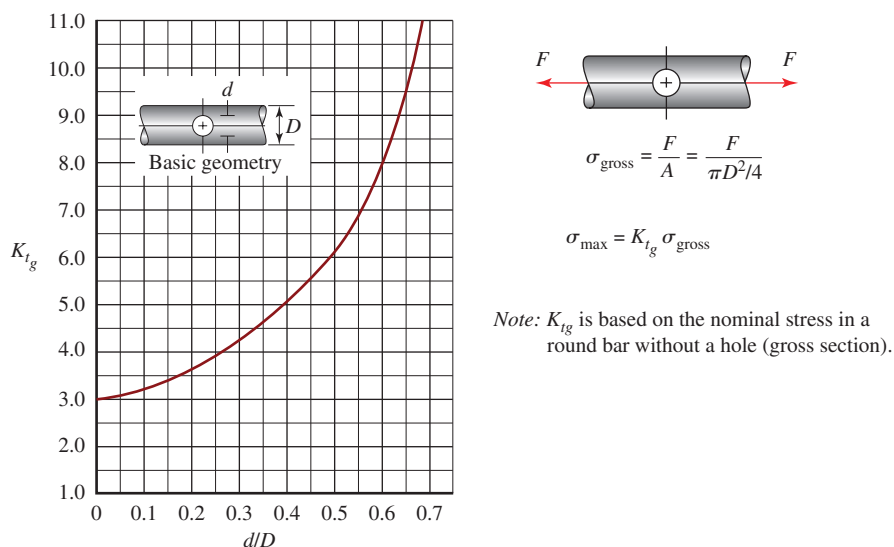


FIGURE A18-3 Round bar with a transverse hole in tension

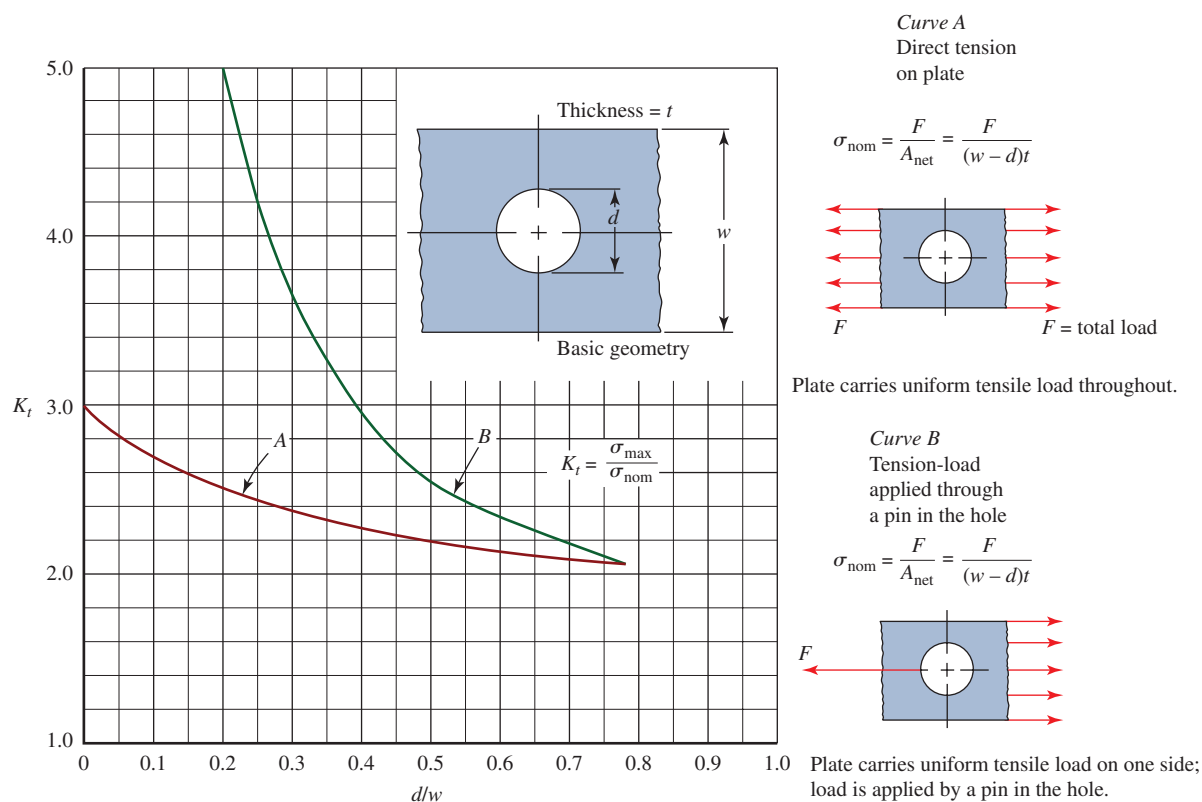


FIGURE A18-4 Flat plate with a central hole in tension

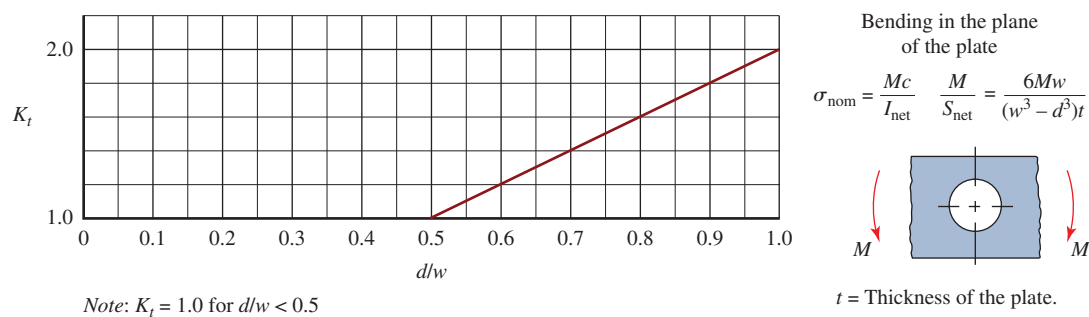


FIGURE A18-5 Flat plate with a central hole in bending

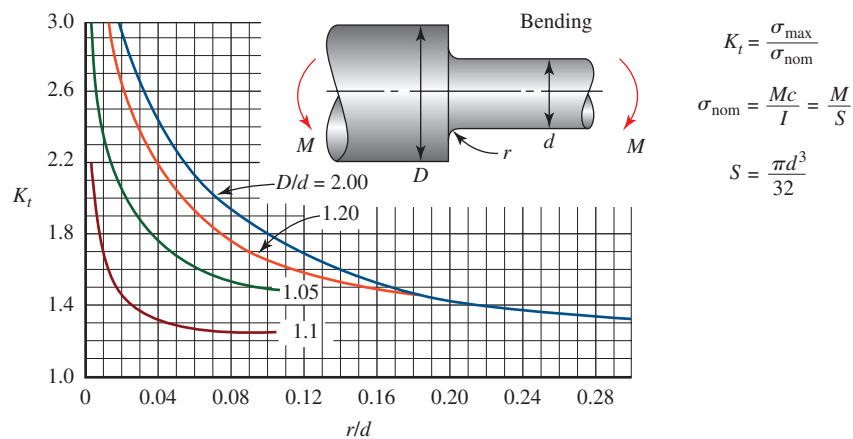


FIGURE A18-6 Stepped round bar in bending

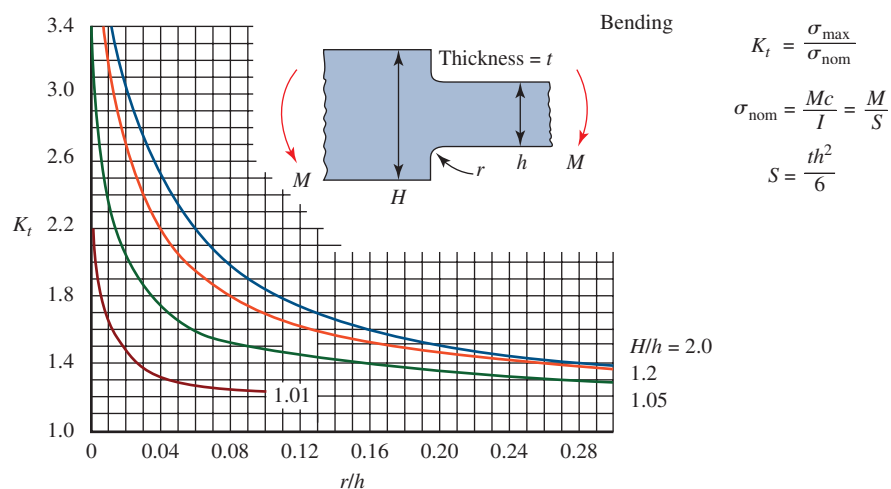


FIGURE A18-7 Stepped flat plate in bending

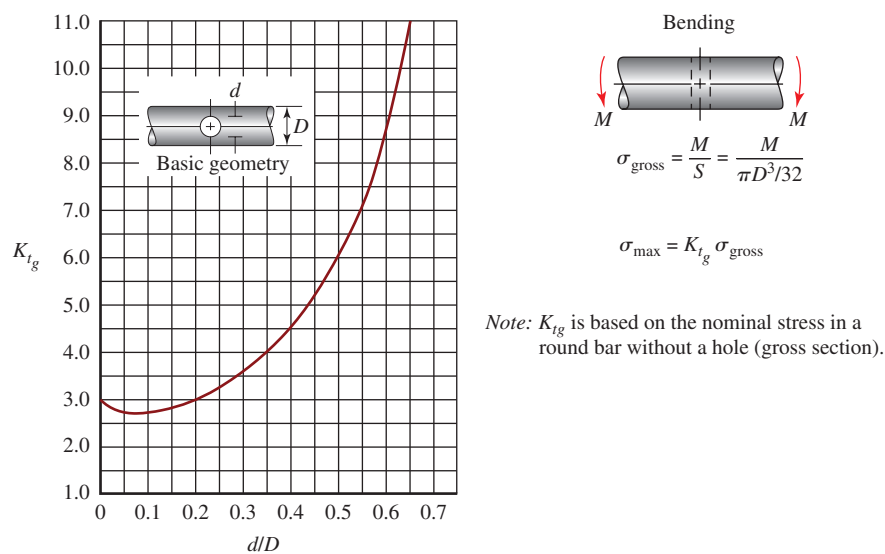


FIGURE A18-8 Round bar with a transverse hole in bending

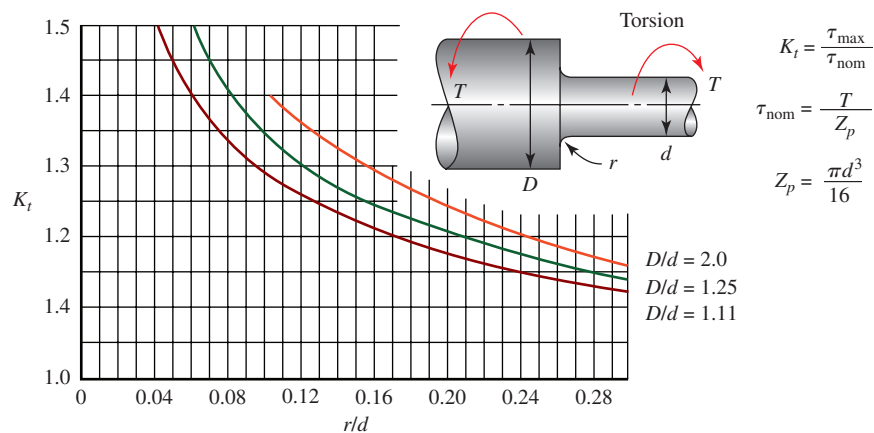


FIGURE A18-9 Stepped round bar in torsion

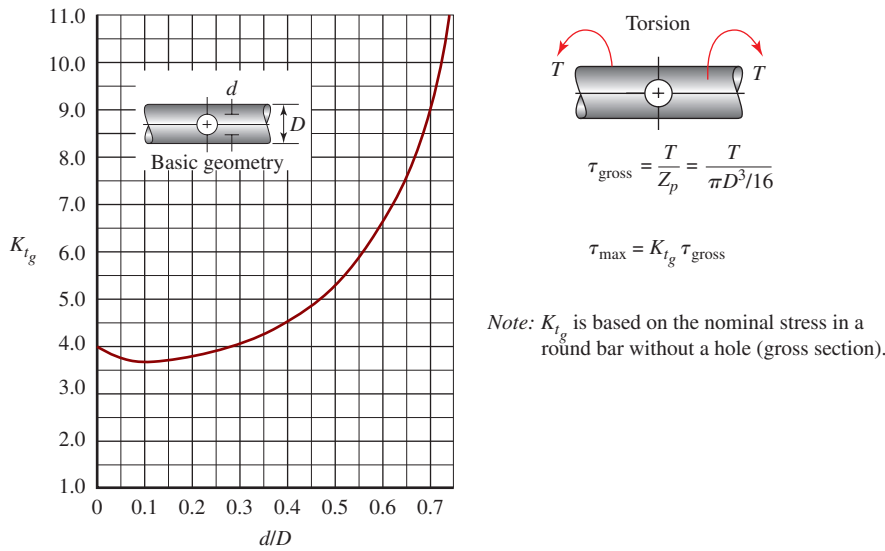


FIGURE A18-10 Round bar with a transverse hole in torsion

APPENDIX 19 Geometry Factor, I , for Pitting for Spur Gears

Section 9-6 introduced the geometry factor, I , for pitting resistance for spur gears as a factor that relates the gear tooth geometry to the radius of curvature of the teeth. The value of I is to be determined at the *lowest point of single tooth contact* (LPSTC). The AGMA defines I as

$$I = C_c C_x$$

where

C_c is the curvature factor at the pitch line

C_x is the factor to adjust for the specific height of the LPSTC

The variables involved are to be the pressure angle, ϕ , the number of teeth in the pinion, N_p , and the gear ratio $m_G = N_G/N_p$. Note that m_G is always greater than or equal to 1.0, regardless of which gear is driving. The value of C_c is easily calculated directly from

$$C_c = \frac{\cos \phi \sin \phi}{2} \frac{m_G}{m_G + 1}$$

The computation of the value of C_x requires the evaluation of several other terms.

$$C_x = \frac{R_1 R_2}{R_P R_G}$$

where each term is developed in the following equations in terms of ϕ , N_p , and m_G along with P_d . It will be shown that the diametral pitch appears in the denominator of each term, and it can then be canceled. We also express each term in the form C/P_d for convenience.

R_P = Radius of curvature for pinion at the pitch point

$$R_P = \frac{D_P \sin \phi}{2} = \frac{N_P \sin \phi}{2 P_d} = \frac{C_1}{P_d}$$

R_G = Radius of curvature for gear at the pitch point

$$R_G = \frac{D_G \sin \phi}{2} = \frac{D_P m_G \sin \phi}{2} = \frac{N_P m_G \sin \phi}{2 P_d} = \frac{C_2}{P_d} = \frac{m_G C_1}{P_d}$$

R_1 = Radius of curvature of the pinion at the LPSTC = $R_P - Z_c$

R_2 = Radius of curvature of the gear at the LPSTC = $R_G + Z_c$

$Z_c = p_b - Z_a$

$$p_b = \text{Base pitch} = \frac{\pi \cos \phi}{P_d} = \frac{C_3}{P_d}$$

$$Z_a = 0.5 \left[\sqrt{D_{op}^2 - D_{bp}^2} - \sqrt{D_p^2 - D_{bp}^2} \right]$$

We now express all of the diameters in this equation in terms of ϕ , N_p , m_G , and P_d .

$$D_{op} = \text{Outside diameter of the pinion} = (N_p + 2)/P_d$$

$$D_p = \text{Pinion diameter} = N_p/P_d$$

$$D_{bp} = \text{Base diameter for the pinion} = D_p \cos \phi = (N_p \cos \phi)/P_d$$

Note that each term has the diametral pitch P_d in the denominator. It can then be factored outside the square root sign. The resulting equation for Z_a is

$$Z_a = \frac{0.5}{P_d} \left[\sqrt{(N_p + 2)^2 - (N_p \cos \phi)^2} - \sqrt{N_p^2 - (N_p \cos \phi)^2} \right] = \frac{C_4}{P_d}$$

Now we can define Z_c .

$$Z_c = p_b - Z_a = \frac{C_3}{P_d} - \frac{C_4}{P_d} = \frac{C_3 - C_4}{P_d}$$

We can now complete the equations for R_1 and R_2 .

$$R_1 = R_p - Z_c = \frac{C_1}{P_d} - \frac{C_3 - C_4}{P_d} = \frac{C_1 - C_3 + C_4}{P_d}$$

$$R_2 = R_G + Z_c = \frac{C_2}{P_d} + \frac{C_3 - C_4}{P_d} = \frac{C_2 + C_3 - C_4}{P_d}$$

Finally, all of these terms can be substituted into the equation for C_x .

$$C_x = \frac{R_1 R_2}{R_p R_G} = \frac{[(C_1 - C_3 + C_4)/P_d] [(C_2 + C_3 - C_4)/P_d]}{(C_1/P_d)(C_2/P_d)}$$

Now you can see that the diametral pitch P_d cancels out, resulting in the final form,

$$C_x = \frac{R_1 R_2}{R_p R_G} = \frac{(C_1 - C_3 + C_4)(C_2 + C_3 - C_4)}{(C_1)(C_2)}$$

The algorithm for computing I can now be stated. First compute each of the C terms.

$$C_1 = (N_p \sin \phi)/2$$

$$C_2 = (N_p m_G \sin \phi)/2 = (C_1)(m_G)$$

$$C_3 = \pi \cos \phi$$

$$C_4 = 0.5 \left[\sqrt{(N_p + 2)^2 - (N_p \cos \phi)^2} - \sqrt{N_p^2 - (N_p \cos \phi)^2} \right]$$

$$C_x = \frac{R_1 R_2}{R_p R_G} = \frac{(C_1 - C_3 + C_4)(C_2 + C_3 - C_4)}{(C_1)(C_2)}$$

$$C_c = \frac{\cos \phi \sin \phi}{2} \frac{m_G}{m_G + 1}$$

Finally, $I = C_c C_x$

**Example Problem
A19-1**

Compute the value for the geometry factor, I , for pitting for the following data: Two spur gears in mesh with a pressure angle of 20° , $N_P = 30$, $N_G = 150$.

Solution

First compute: $m_G = N_G/N_P = 150/30 = 5.0$

$$\text{Then } C_1 = (N_P \sin \phi)/2 = (30)(\sin 20^\circ)/2 = 5.1303$$

$$C_2 = (N_P m_G \sin \phi)/2 = (C_1)(m_G) = 25.652$$

$$C_3 = \pi \cos \phi = \pi \cos 20^\circ = 2.9521$$

$$C_4 = 0.5 \left[\sqrt{(N_P + 2)^2 - (N_P \cos \phi)^2} - \sqrt{N_P^2 - (N_P \cos \phi)^2} \right]$$

$$C_4 = 0.5 \left[\sqrt{(30 + 2)^2 - (30 \cos 20^\circ)^2} - \sqrt{30^2 - (30 \cos 20^\circ)^2} \right] = 2.4407$$

$$C_x = \frac{R_1 R_2}{R_P R_G} = \frac{(C_1 - C_3 + C_4)(C_2 + C_3 - C_4)}{(C_1)(C_2)}$$

$$C_x = \frac{(5.1303 - 2.9521 + 2.4407)(25.652 + 2.9521 - 2.4407)}{(5.1303)(25.652)} = 0.91826$$

$$C_c = \frac{\cos \phi \sin \phi}{2} \frac{m_G}{m_G + 1} = \frac{\cos 20^\circ \sin 20^\circ (5)}{2(5 + 1)} = 0.13391$$

$$\text{Finally, } I = C_c C_x = (0.13391)(0.91826) = 0.12297 \approx 0.123$$

This process lends itself well to programming in a spreadsheet, MATLAB, BASIC, or any other convenient computational aid.

ANSWERS TO SELECTED PROBLEMS

Given here are the answers to problems for which there are unique solutions. Many of the problems in this book are true design problems, and individual design decisions are required to arrive at the solutions. Others are of the review question form for which the answers are in the text of the associated chapter. It should also be noted that some of the problems require the selection of design factors and the use of data from charts and graphs. Because of the judgment and interpolation required, some of the answers may be slightly different from your solutions.

Chapter 1 The Nature of Mechanical Design

15. $D = 44.5 \text{ mm}$
16. $L = 14.0 \text{ m}$
17. $T = 1418 \text{ N} \cdot \text{m}$
18. $A = 2658 \text{ mm}^2$
19. $S = 2.43 \times 10^5 \text{ mm}^3$
20. $I = 3.66 \times 10^7 \text{ mm}^4$
21. From Table A15-3:
Angle $50 \times 100 \times 6$ or $75 \times 75 \times 6$
Both have $A = 864 \text{ mm}^2$
22. $P = 5.59 \text{ kW}$
23. $s_u = 876 \text{ MPa}$
24. Weight = 48.9 N
25. $T = 20.3 \text{ N} \cdot \text{m}$
 $\theta = 0.611 \text{ rad}$
Scale = $5.14 \text{ lb} \cdot \text{in/deg}$
Scale = $33.3 \text{ N} \cdot \text{m/rad}$
26. Energy = $1.03 \times 10^{11} \text{ lb} \cdot \text{ft/yr}$
Energy = $3.88 \times 10^7 \text{ W} \cdot \text{h/yr}$
27. $\mu = 540 \text{ lb} \cdot \text{s/ft}^2$; $\mu = 25.9 \times 10^3 \text{ N} \cdot \text{s/m}^2$
28. $4.60 \times 10^9 \text{ rev}$

Chapter 2 Materials in Mechanical Design

9. No. The percent elongation must be greater than 5.0 percent to be ductile.
11. $G = 42.9 \text{ GPa}$
12. Hardness = 52.8 HRC
13. Tensile strength = 235 ksi (Approximately)

Questions 14–17 ask what is wrong with the given statements.

14. Annealed steels typically have hardness values in the range from 120 HB to 200 HB. A hardness of 750 HB is extremely hard and characteristic of as-quenched high alloy steels.
15. The HRC scale is normally limited to HRC 100.
16. The HRC hardness is normally no lower than HRC 20.
17. The given relationship between hardness and tensile strength is only valid for steels.
18. Charpy and Izod
19. Iron and carbon. Manganese and other elements are often present.
20. Iron, carbon, manganese, nickel, chromium, molybdenum.
21. Approximately 0.40 percent.
22. Low carbon: Less than 0.30 percent
Medium carbon: 0.30 to 0.50 percent
High carbon: 0.50 to 0.95 percent
23. Nominally 1.0 percent.
24. SAE 12L13 steel has lead added to improve machinability.
25. SAE 1045, 4140, 4640, 5150, 6150, 8650.
26. SAE 1045, 4140, 4340, 4640, 5150, 6150, 8650.
27. Wear resistance, strength, ductility. SAE 1080.
28. SAE 5160 OQT 1000 is a chromium steel, having nominally 0.80 percent chromium and 0.60 percent carbon, a high carbon alloy steel. It has fairly high strength and good ductility. It was through-hardened, quenched in oil, and tempered at 1000°F.

29. Yes, with careful specification of the quenching medium. A hardness of HRC 40 is equivalent to HB 375. Appendix 3 indicates that oil quenching would not produce an adequate hardness. However, Figure A4–1 shows that a hardness of HB 400 could be obtained by quenching in water and tempering in 700°F while still having 20% elongation for good ductility.
33. SAE 200 and 300 series
34. Chromium
35. ASTM A992 structural steel
37. Possible answers: Gray iron, malleable iron, ductile iron, austempered ductile iron, carbide austempered ductile iron, white iron
41. Stamping dies, punches, gages
43. Strain hardened
46. Alloy 6061
48. Bronze is an alloy of copper with several alloying elements.
50. Gears and bearings
55. Thermoset Plastics: Polyesters, epoxies, polyimides, phenolics
Thermoplastics: Polyethylenes, polyamides, polycarbonates, polyvinyl chlorides
56. Glass, carbon, graphite, boron, aramid, silicon carbide
60. Auto and truck bodies; large housings

Supplementary Problems

1. Poisson's ratio:
 - (a) Carbon steel—0.29;
 - (c) Lead—0.43;
 - (e) Concrete—0.10 to 0.25
3. Erosive, abrasive, adhesive, fretting, surface fatigue
5. SAE 304 and SAE 316
9. (a) DIN 42CrMo4 or W-1.7225;
(b) BS 708A42;
(c) EN 42CrMo4;
(d) GB ML42CrMo4;
(e) JIS SCM 440H
12. (a) DIN AlZnMgCu1.5 or W-3.4365;
(b) BS L.95, L.96;
(c) EN AlZn6MgCu
13. Water, brine, mineral oil, water-soluble polyalkylene glycol (PAG)
15. ASTM A27/A27M; A915/A915M; A128/A128M; A148/A148M
17. Carbide austempered ductile iron, used for: railroad rolling stock, earthmoving equipment, agricultural machinery, crushers
23. Carbon steel F-0008-HT, $s_u = 85$ ksi (590 MPa);
Low-alloy steel FL-4405-HT, $s_u = 160$ ksi (1100 MPa);
Diffusion-alloyed steel FD-0205-HT, $s_u = 130$ ksi (900 MPa);
Sinter-hardened steel FLC-4608-HT, $s_u = 100$ ksi (690 MPa)
27. Aluminum 2014, 2024, 6061
31. (a) Bearing bronze C93200l;
(c) Muntz metal C37000;
(e) Copper-nickel-zinc alloy C96200
35. Metals, polymers, ceramics, glasses, elastomers, hybrids
37. Metals, ceramics, composites, polymers, wood, rubbers/elastomers, foams

Chapter 3 Stress and Deformation Analysis

1. $\sigma = 31.8$ MPa; $\delta = 0.12$ mm
2. $\sigma = 44.6$ MPa
3. $\sigma = 66.7$ MPa
4. $\sigma = 5375$ psi
5. $\sigma = 17\,200$ psi
6. For all materials, $\sigma = 34.7$ MPa
Deflection:
 - (a) $\delta = 0.277$ mm
 - (b) $\delta = 0.277$ mm
 - (c) $\delta = 0.347$ mm
 - (d) $\delta = 0.830$ mm
 - (e) $\delta = 0.503$ mm
 - (f) $\delta = 27.7$ mm
 - (g) $\delta = 7.56$ mm.

Note: The stress is close to the ultimate strength for (f) and (g).
7. Force = 2556 lb; $\sigma = 2506$ psi
8. $\sigma = 595$ psi
9. Force = 1061 lb
10. $D = 0.274$ in
13. $\sigma_{AD} = \sigma_{DE} = 6198$ psi
 $\sigma_{EF} = 7748$ psi
 $\sigma_{BD} = 0$ psi
 $\sigma_{BE} = 5165$ psi
 $\sigma_{AB} = \sigma_{CE} = -4132$ psi
 $\sigma_{BC} = -3099$ psi
 $\sigma_{CF} = -5165$ psi
15. $\sigma = 144$ MPa
16. At pins A and C: $\tau_A = \tau_C = 7958$ psi
At pin B: $\tau_B = 10\,190$ psi

19. $\tau = 98.8 \text{ MPa}$
 21. $\tau = 547 \text{ MPa}$
 22. $\tau = 32.6 \text{ MPa}$
 23. $\theta = 0.79^\circ$
 25. $\tau = 32\,270 \text{ psi}$
 28. $\tau = 70.8 \text{ MPa}$; $\theta = 1.94^\circ$
 30. $T = 9624 \text{ lb} \cdot \text{in}$; $\theta = 1.83^\circ$
 31. Required section modulus $= 2.40 \text{ in}^3$. Standard nominal sizes given for each shape:
 (a) Each side $= 2.50 \text{ in}$
 (c) Width $= 5.00 \text{ in}$; height $= 1.75 \text{ in}$
 (e) S4 $\times$ 7.7
 (g) 4-in schedule 40 pipe
 32. Weights:
 (a) 212 lb
 (c) 297 lb
 (e) 77.0 lb
 (g) 107.9 lb
 33. Maximum deflection Deflection at loads
 (a) 0.701 in 0.572 in
 (c) 1.021 in 0.836 in
 (e) 0.375 in 0.307 in
 (g) 0.315 in 0.258 in
 34. $M_A = 330 \text{ N} \cdot \text{m}$; $M_B = 294 \text{ N} \cdot \text{m}$;
 $M_C = -40 \text{ N} \cdot \text{m}$
 36. (a) $y_A = 0.238 \text{ in}$; $y_B = 0.688 \text{ in}$
 (b) $y_A = 0.047 \text{ in}$; $y_B = 0.042 \text{ in}$
 38. $\sigma = 3480 \text{ psi}$; $\tau = 172 \text{ psi}$
 For Problems 39 through 49, complete solutions require drawings. Listed below are the maximum bending moments only:
 39. $480 \text{ lb} \cdot \text{in}$
 41. $120 \text{ lb} \cdot \text{in}$
 43. $93\,750 \text{ N} \cdot \text{mm}$
 45. $2940 \text{ lb} \cdot \text{in}$
 47. $11\,811 \text{ lb} \cdot \text{in}$
 49. $8640 \text{ lb} \cdot \text{in}$
 51. $\sigma = 62.07 \text{ MPa}$
 53. (a) $\sigma = 20.94 \text{ MPa}$ tension on top of lever
 (b) At section B, $h = 35.1 \text{ mm}$; at C, $h = 18 \text{ mm}$
 55. $\sigma = 84.6 \text{ MPa}$ tension
 57. Sides $= 0.50 \text{ in}$
 59. Maximum $\sigma = -1.42 \text{ MPa}$ compression on the top surface between A and C
 61. $\sigma = 89.7 \text{ MPa}$

63. Left: $\sigma = 39\,206 \text{ psi}$
 Middle: $\sigma = 29\,512 \text{ psi}$
 Right: $\sigma = 31\,551 \text{ psi}$
 65. $\sigma = 98.6 \text{ MPa}$
 67. $\sigma = 32\,564 \text{ psi}$
 69. Tension in member A–B: $\sigma = 50.0 \text{ MPa}$
 Shear in pin: $\tau = 199 \text{ MPa}$
 Bending in A–C at B: $\sigma = 14\,063 \text{ MPa}$ (Very high; redesign)
 71. $\sigma = 186 \text{ MPa}$
 73. $\sigma_{\max} = 121.7 \text{ MPa}$ at first step, 40 mm from either support.
 75. $\sigma = 108.8 \text{ MPa}$
 77. With pivot at top hole: $\sigma_{\max} = 10\,400 \text{ psi}$ at fulcrum
 With pivot at bottom hole: $\sigma_{\max} = 5600 \text{ psi}$ at fulcrum
 81. $F = 126 \text{ N}$
 83. $N = 0.651$ —Inside surface;
 $N = 0.913$ —Outside surface
 Both indicate failure by yielding.
 85. $N = 3.12$ —Inside surface; $N = 3.65$ —Outside surface

Chapter 4 Combined Stresses and Mohr's Circle

	Maximum Principal Stress	Minimum Principal Stress	Maximum Shear Stress
1.	24.14 ksi	−4.14 ksi	14.14 ksi
3.	50.0 ksi	−50.0 ksi	50.0 ksi
5.	42.5 ksi	−122.5 ksi	82.5 ksi
7.	71.0 ksi	−51.0 ksi	61.0 ksi
9.	44.3 MPa	−144.3 MPa	94.3 MPa
11.	61.3 MPa	−91.3 MPa	76.3 MPa
13.	86.8 MPa	−156.8 MPa	121.8 MPa
15.	250.0 MPa	−80.0 MPa	165.0 MPa
17.	453 MPa	−353 MPa	403 MPa
19.	42.2 MPa	−52.2 MPa	47.2 MPa
21.	40.0 ksi	0 ksi	20.0 ksi
23.	42.8 ksi	−29.8 ksi	36.3 ksi
25.	23.9 ksi	−1.9 ksi	12.9 ksi
27.	328 MPa	0 MPa	164 MPa
29.	0 kPa	−868 kPa	434 kPa
31.	26.24 ksi	−5.70 ksi	15.97 ksi
33.	7730 psi	−4.0 psi	3867 psi
35.	398 psi	−6366 psi	3382 psi

Chapter 5 Design for Different Types of Loading

Stress Ratio

1. $\sigma_{\max} = 44.6 \text{ MPa}$
 $\sigma_{\min} = 6.37 \text{ MPa}$
 $\sigma_m = 25.5 \text{ MPa}$
 $\sigma_a = 19.1 \text{ MPa}$
 $R = 0.143$
3. $\sigma_{\max} = 5375 \text{ psi}$
 $\sigma_{\min} = -750 \text{ psi}$
 $\sigma_m = 2313 \text{ psi}$
 $\sigma_a = 3063 \text{ psi}$
 $R = -0.140$
5. $\sigma_{\max} = 110.3 \text{ MPa}$
 $\sigma_{\min} = 50.9 \text{ MPa}$
 $\sigma_m = 80.6 \text{ MPa}$
 $\sigma_a = 29.7 \text{ MPa}$
 $R = 0.462$
7. $\sigma_{\max} = 9868 \text{ psi}$
 $\sigma_{\min} = 1645 \text{ psi}$
 $\sigma_m = 5757 \text{ psi}$
 $\sigma_a = 4112 \text{ psi}$
 $R = 0.167$
9. $\sigma_{\max} = 475 \text{ MPa}$
 $\sigma_{\min} = 297 \text{ MPa}$
 $\sigma_m = 386 \text{ MPa}$
 $\sigma_a = 89 \text{ MPa}$
 $R = 0.625$

Endurance Strength

11. $s'_n = 211 \text{ MPa}$
13. $s'_n = 31.2 \text{ ksi}$

Design and Analysis

19. $N = 1.57$ (low)
23. $N = 2.82$ at right load. OK
25. $N = 11.8$
29. $D = 40 \text{ mm}$; $a = 9.93 \text{ mm}$
31. $2\frac{1}{2}$ in Schedule 40 steel pipe or
 $2\frac{1}{2}$ in steel tube with 5 gauge wall ($t = 0.220 \text{ in}$)
 Steel tube is lighter with smaller area.
33. $b = 1.80 \text{ in}$
35. $N = 9.11$
36. $\sigma = 34.7 \text{ MPa}$
 - (a) $N = 5.96$
 - (c) $N = 7.95$
 - (e) $N = 23.8$
 - (g) $N = 1.30$ (low)

37. $N = 3.37$
39. Force = 1061 lb; $D = 5/16 \text{ in}$
41. $N = 1.64$ (low)
43. $N = 5.10$
45. $D = 1.75 \text{ in}$
49. $N = 7.92$
51. $N = 3.19$
55. Specify: ASTM A536 Grade 100-70-03
57. $N = 1.61$ (low)
59. One possible material: SAE 3140 OQT 1300
 $s_u = 115 \text{ ksi}$, $s_y = 94 \text{ ksi}$, 23% elongation
61. Design a
64. $N = 0.24$ at pin holes (Failure)
67. $N = 1.65$ at B (low)
69. $N = 1.65$ (low)
73. $N = 1.29$ (low)
75. $r_{\min} = 0.22 \text{ in}$

Chapter 6 Columns

1. $P_{cr} = 4473 \text{ lb}$
2. $P_{cr} = 14\,373 \text{ lb}$
5. $P_{cr} = 4473 \text{ lb}$
6. $P_{cr} = 32.8 \text{ lb}$
8. (a) Pinned ends: $P_{cr} = 7498 \text{ lb}$
 (b) Fixed ends: $P_{cr} = 12\,000 \text{ lb}$
 (c) Fixed-pinned ends: $P_{cr} = 10\,300 \text{ lb}$
 (d) Fixed-free ends: $P_{cr} = 1700 \text{ lb}$
10. $D = 1.45 \text{ in}$ required. Use $D = 1.50 \text{ in}$.
14. $S = 1.423 \text{ in}$ required. Use $S = 1.500 \text{ in}$.
16. Use $D = 1.50 \text{ in}$.
23. $P = 1189 \text{ lb}$
25. $P = 1877 \text{ lb}$
27. $\sigma = 212 \text{ MPa}$; $y = 25.7 \text{ mm}$
 The stress is above the yield strength of the column material. Specify a larger pipe size.
28. $\sigma = 6685 \text{ psi}$; $y = 0.045 \text{ in}$
30. $P = 37\,500 \text{ psi}$
31. $P_a = 22\,600 \text{ lb}$
33. $4 \times 4 \times 1/2$ is the smallest outside dimension.
 $I = 11.9 \text{ in}^4$. Weight = 21.5 lb/ft.
 $6 \times 4 \times 1/4$ is the lightest from those listed.
 $I_{\min} = I_y = 11.1 \text{ in}^4$. Weight = 15.6 lb/ft.
35. $P_a = 11\,750 \text{ lb}$
37. $P_a = 18\,300 \text{ lb}$
39. Piston rod is safe.

Chapter 7 Belt Drives and Chain Drives

V-Belt Drives

1. 3V Belt, 75 inches long
2. $C = 22.00$ in
3. $\theta_1 = 157^\circ$; $\theta_2 = 203^\circ$
10. $v_b = 2405$ ft/min
13. $P = 6.05$ hp

Roller Chain

25. No. 80 chain
28. Design power rating = 18.08 hp; Type B lubrication (bath)
29. Design power rating = 45.2 hp
34. $L = 96$ in, 128 pitches
35. $C = 35.57$ in

Chapter 8 Kinematics of Gears

Gear Geometry

1. $N = 44$; $P_d = 12$
 - (a) $D = 3.667$ in
 - (b) $p = 0.2618$ in
 - (c) $m = 2.117$ mm
 - (d) $m = 2.00$ mm
 - (e) $a = 0.0833$ in
 - (f) $b = 0.1042$ in
 - (g) $c = 0.0208$ in
 - (h) $h_t = 0.1875$ in
 - (i) $h_k = 0.1667$ in
 - (j) $t = 0.131$ in
 - (k) $D_o = 3.833$ in
3. $N = 45$; $P_d = 2$
 - (a) $D = 22.500$ in
 - (b) $p = 1.571$ in
 - (c) $m = 12.70$ mm
 - (d) $m = 12.0$ mm
 - (e) $a = 0.5000$ in
 - (f) $b = 0.6250$ in
 - (g) $c = 0.1250$ in
 - (h) $h_t = 1.1250$ in
 - (i) $h_k = 1.0000$ in
 - (j) $t = 0.7854$ in
 - (k) $D_o = 23.500$ in
5. $N = 22$; $P_d = 1.75$
 - (a) $D = 12.571$ in
 - (b) $p = 1.795$ in
 - (c) $m = 14.514$ mm
 - (d) $m = 16.0$ mm
 - (e) $a = 0.5714$ in
 - (f) $b = 0.7143$ in
 - (g) $c = 0.1429$ in
 - (h) $h_t = 1.2857$ in
 - (i) $h_k = 1.1429$ in
 - (j) $t = 0.8976$ in
 - (k) $D_o = 13.714$ in
7. $N = 180$; $P_d = 80$
 - (a) $D = 2.2500$ in
 - (b) $p = 0.0393$ in
 - (c) $m = 0.318$ mm
 - (d) $m = 0.30$ mm
 - (e) $a = 0.0125$ in
 - (f) $b = 0.0170$ in
 - (g) $c = 0.0045$ in
 - (h) $h_t = 0.0295$ in
 - (i) $h_k = 0.0250$ in
 - (j) $t = 0.0196$ in
 - (k) $D_o = 2.2750$ in
9. $N = 28$; $P_d = 20$
 - (a) $D = 1.4000$ in
 - (b) $p = 0.1571$ in
 - (c) $m = 1.270$ mm
 - (d) $m = 1.25$ mm
 - (e) $a = 0.0500$ in
 - (f) $b = 0.0620$ in
 - (g) $c = 0.0120$ in
 - (h) $h_t = 0.1120$ in
 - (i) $h_k = 0.1000$ in
 - (j) $t = 0.0785$ in
 - (k) $D_o = 1.5000$ in
11. $N = 45$; $m = 1.25$
 - (a) $D = 56.250$ mm
 - (b) $p = 3.927$ mm
 - (c) $P_d = 20.3$
 - (d) $P_d = 20$
 - (e) $a = 1.25$ mm
 - (f) $b = 1.563$ mm
 - (g) $c = 0.313$ mm
 - (h) $h_t = 2.813$ mm
 - (i) $h_k = 2.500$ mm
 - (j) $t = 1.963$ mm
 - (k) $D_o = 58.750$ mm

13. $N = 22; m = 20$

- (a) $D = 440.00 \text{ mm}$
- (b) $p = 62.83 \text{ mm}$
- (c) $P_d = 1.270$
- (d) $P_d = 1.25$
- (e) $a = 20.0 \text{ mm}$
- (f) $b = 25.00 \text{ mm}$
- (g) $c = 5.000 \text{ mm}$
- (h) $h_t = 45.00 \text{ mm}$
- (i) $h_k = 40.00 \text{ mm}$
- (j) $t = 31.42 \text{ mm}$
- (k) $D_o = 480.00 \text{ mm}$

15. $N = 180; m = 0.4$

- (a) $D = 72.00 \text{ mm}$
- (b) $p = 1.26 \text{ mm}$
- (c) $P_d = 63.5$
- (d) $P_d = 64$
- (e) $a = 0.40 \text{ mm}$
- (f) $b = 0.500 \text{ mm}$
- (g) $c = 0.100 \text{ mm}$
- (h) $h_t = 0.90 \text{ mm}$
- (i) $h_k = 0.80 \text{ mm}$
- (j) $t = 0.628 \text{ mm}$
- (k) $D_o = 72.80 \text{ mm}$

17. $N = 28; m = 0.8$

- (a) $D = 22.40 \text{ mm}$
- (b) $p = 2.51 \text{ mm}$
- (c) $P_d = 31.75$
- (d) $P_d = 32$
- (e) $a = 0.80 \text{ mm}$
- (f) $b = 1.000 \text{ mm}$
- (g) $c = 0.200 \text{ mm}$
- (h) $h_t = 1.800 \text{ mm}$
- (i) $h_k = 1.60 \text{ mm}$
- (j) $t = 1.257 \text{ mm}$
- (k) $D_o = 24.00 \text{ mm}$

19. Problem 1: $P_d = 12$; backlash = 0.006 to 0.009 in
 Problem 12: $m = 12$; backlash = 0.52 to 0.82 mm

21.(a) $C = 14.000 \text{ in}$

- (b) $VR = 4.600$
- (c) $n_G = 48.9 \text{ rpm}$
- (d) $v_t = 294.5 \text{ ft/min}$

23. (a) $C = 2.266 \text{ in}$

- (b) $VR = 6.25$
- (c) $n_G = 552 \text{ rpm}$
- (d) $v_t = 565 \text{ ft/min}$

25.(a) $C = 90.00 \text{ mm}$

- (b) $VR = 3.091$
- (c) $n_G = 566 \text{ rpm}$
- (d) $v_t = 4.03 \text{ m/s}$

27. (a) $C = 162.0 \text{ mm}$

- (b) $VR = 1.250$
- (c) $n_G = 120 \text{ rpm}$
- (d) $v_t = 1.13 \text{ m/s}$

For problems 29 through 32, the following are errors in the given statements:

29. The pinion and the gear cannot have different pitches.

30. The actual center distance should be 8.333 in.

31. There are too few teeth in the pinion; interference is to be expected.

32. The actual center distance should be 2.156 in. Apparently, the outside diameters were used instead of the pitch diameters to compute C .

33. $Y = 8.45 \text{ in}; X = 10.70 \text{ in}$

35. $Y = 44.00 \text{ mm}; X = 58.40 \text{ mm}$

37. Output speed = 111 rpm CCW

39. Output speed = 144 rpm CW

Helical Gearing

41. $p = 0.3927 \text{ in}$ $p_n = 0.3401 \text{ in}$
 $P_{nd} = 9.238$ $P_x = 0.680 \text{ in}$
 $D = 5.625 \text{ in}$ $\phi_n = 12.62^\circ$
 $F/P_x = 2.94$ axial pitches in the face width

42. $P_d = 8.485$ $p = 0.370 \text{ in}$
 $p_c = 0.2618 \text{ in}$ $P_x = 0.370 \text{ in}$
 $\phi_t = 27.2^\circ$ $D = 5.657 \text{ in}$
 $F/P_x = 4.05$ axial pitches in the face width

Bevel Gears

45. Selected results: $F = 1.25 \text{ in}$ specified.

$d = 2.500 \text{ in}$	$D = 7.500 \text{ in}$
$\gamma = 18.435^\circ$	$\Gamma = 71.565^\circ$
$A_o = 3.953 \text{ in}$	$F_{\text{nom}} = 1.186 \text{ in}$
$A_m = A_{mG} = 3.328 \text{ in}$	$b = 0.281 \text{ in}$
$c = 0.035 \text{ in}$	$h_m = 0.316 \text{ in}$
$a_p = 0.213 \text{ in}$	$a_G = 0.068 \text{ in}$
$d_o = 2.992 \text{ in}$	$D_o = 7.555 \text{ in}$

49. Selected results: $F = 0.800 \text{ in}$ specified.

$d = 1.500 \text{ in}$	$D = 6.000 \text{ in}$
$\gamma = 14.03^\circ$	$\Gamma = 75.97^\circ$
$A_o = 3.092 \text{ in}$	$F_{\text{nom}} = 0.928 \text{ in}$
$A_m = A_{mG} = 2.692 \text{ in}$	$b = 0.145 \text{ in}$
$c = 0.018 \text{ in}$	$h_m = 0.163 \text{ in}$
$a_p = 0.112 \text{ in}$	$a_G = 0.033 \text{ in}$
$d_o = 1.755 \text{ in}$	$D_o = 6.020 \text{ in}$

Wormgearing

52. $L = 0.3142$ in $\lambda = 4.57^\circ$
 $a = 0.100$ in $b = 0.1157$ in
 $D_{OW} = 1.450$ in $D_{RW} = 1.0186$ in
 $D_G = 4.000$ in $C = 2.625$ in
 $VR = 40$

Analysis of Complex Gear Trains

59. 0.4067 rpm
 61. 0.5074 rpm

Kinematic Design of a Single Gear Pair

63. $N_P = 22$, $N_G = 38$
 65. $N_P = 19$, $N_G = 141$

Kinematic Design of Gear Trains

68. One possible solution: Triple reduction;
 Layout as in Figure 8-48.
 $N_A = N_C = N_E = 17$, $N_B = 136$, $N_D = 119$,
 $N_F = 85$
 $TV = 280$ exactly; $n_{out} = 12$ rpm exactly;
 Used factoring method
69. One possible solution: Triple reduction with an idler
 $N_{P1} = 18$, $N_{G1} = 126$, $N_{P2} = 18$, $N_{G2} = 108$,
 $N_{P3} = 18$, $N_{G3} = 135$, $N_{idler} = 18$
 $n_{out} = 13.33$ rpm
72. One possible solution: Double reduction;
 Layout as in Figure 8-47.
 $N_A = N_C = 18$, $N_B = 75$, $N_D = 51$
 $n_{out} = 148.2$ rpm

Chapter 9 Spur Gear Design

1. (a) $n_G = 486.1$ rpm
 (b) $VR = m_G = 3.600$
 (c) $D_P = 1.667$ in; $D_G = 6.000$ in
 (d) $C = 3.833$ in
 (e) $v_t = 764$ ft/min
 (f) $T_P = 270$ lb·in; $T_G = 972$ lb·in
 (g) $W_t = 324$ lb
 (h) $W_r = 118$ lb
 (i) $W_N = 345$ lb
3. (a) $n_G = 752.7$ rpm
 (b) $VR = m_G = 4.583$
 (c) $D_P = 1.000$ in; $D_G = 4.583$ in
 (d) $C = 2.792$ in
 (e) $v_t = 903$ ft/min
 (f) $T_P = 13.7$ lb·in; $T_G = 62.8$ lb·in
 (g) $W_t = 27.4$ lb
 (h) $W_r = 10.0$ lb
 (i) $W_N = 29.2$ lb

5. (a) $n_G = 304.4$ rpm
 (b) $VR = m_G = 3.778$
 (c) $D_P = 3.600$ in; $D_G = 13.600$ in
 (d) $C = 8.600$ in
 (e) $v_t = 1084$ ft/min
 (f) $T_P = 2739$ lb·in; $T_G = 10\,348$ lb·in
 (g) $W_t = 1522$ lb
 (h) $W_r = 710$ lb
 (i) $W_N = 1680$ lb
8. A10
 9. A7
 11. A2
 15. A8
26. (a) $s_{at} = 28.26$ ksi; $s_{ac} = 93.50$ ksi – U.S.
 $s_{at} = 194.9$ MPa; $s_{ac} = 644.6$ MPa – SI
 (c) $s_{at} = 43.72$ ksi; $s_{ac} = 157.9$ ksi – U.S.
 $s_{at} = 301.5$ MPa; $s_{ac} = 1088.6$ MPa – SI
 (e) $s_{at} = 36.80$ ksi; $s_{ac} = 104.1$ ksi – U.S.
 $s_{at} = 253.7$ MPa; $s_{ac} = 718.5$ MPa – SI
 (g) $s_{at} = 57.20$ ksi; $s_{ac} = 173.9$ ksi – U.S.
 $s_{at} = 394.3$ MPa; $s_{ac} = 1200.5$ MPa – SI
27. $HB = 300$ for Grade 1; $HB = 192$ for Grade 2
33. (a) $s_{at} = 45.0$ ksi; $s_{ac} = 170.0$ ksi – U.S.
 $s_{at} = 310$ MPa; $s_{ac} = 1172$ MPa – SI
 (c) $s_{at} = 55.0$ ksi; $s_{ac} = 180.0$ ksi – U.S.
 $s_{at} = 379$ MPa; $s_{ac} = 1241$ MPa – SI
 (e) $s_{at} = 55.0$ ksi; $s_{ac} = 180.0$ ksi – U.S.
 $s_{at} = 379$ MPa; $s_{ac} = 1241$ MPa – SI
 (f) $s_{at} = 5.00$ ksi; $s_{ac} = 50.0$ ksi – U.S.
 $s_{at} = 35.0$ MPa; $s_{ac} = 345$ MPa – SI
 (h) $s_{at} = 27.0$ ksi; $s_{ac} = 92.0$ ksi – U.S.
 $s_{at} = 186$ MPa; $s_{ac} = 634$ MPa – SI
 (j) $s_{at} = 23.6$ ksi; $s_{ac} = 65.0$ ksi – U.S.
 $s_{at} = 163$ MPa; $s_{ac} = 448$ MPa – SI
 (l) $s_{at} = 9.00$ ksi; s_{ac} not listed
 $s_{at} = 62.0$ MPa; s_{ac} not listed

34. $h_e = 0.027$ in

35. $h_e = 0.90$ mm

The following three sets of answers are given in groups of four problems that all relate to the same basic set of design data.

Group A

37. $s_{tP} = 32\,740$ psi; $s_{tG} = 26\,940$ psi
 43. $s_{atP} = 34\,460$ psi; $s_{atG} = 28\,100$ psi
 49. $s_{cP} = 172\,100$ psi; $s_{cG} = 172\,100$ psi
 55. $s_{acP} = 189\,200$ psi; $s_{acG} = 185\,100$ psi

Group B

39. $s_{tP} = 2300$ psi; $s_{tG} = 2000$ psi
 45. $s_{atP} = 3700$ psi; $s_{atG} = 3100$ psi
 51. $s_{cP} = 37\,800$ psi; $s_{cG} = 37\,800$ psi
 57. $s_{acP} = 63\,600$ psi; $s_{acG} = 62\,200$ psi

Group C

41. $s_{tP} = 9458$ psi; $s_{tG} = 8134$ psi
 47. $s_{atP} = 10\,254$ psi; $s_{atG} = 8642$ psi
 53. $s_{cP} = 78\,263$ psi; $s_{cG} = 78\,263$ psi
 59. $s_{acP} = 87\,531$ psi; $s_{acG} = 85\,727$ psi

Problems 60–70 are design problems for which there are no unique solutions.

71. Power capacity = 12.9 hp based on gear contact stress at 15 000 h life.

Problems 73–83 are design problems for which there are no unique solutions.

Chapter 10 Helical Gears, Bevel Gears, and Wormgearing

1. $W_t = 89.6$ lb; $W_x = 51.7$ lb; $W_r = 23.2$ lb
 $A_v = 11$; $s_t = 2778$ psi; $s_c = 36\,228$ psi
 Cast iron Class 20
3. $W_t = 143$ lb; $W_x = 143$ lb; $W_r = 37.0$ lb
 $A_v = 9$; $s_t = 9720$ psi; $s_c = 73\,300$ psi
 Ductile iron 60–40–18 or Cast iron Class 40
14. $W_{tP} = W_{tG} = 599$ lb; $W_{xP} = W_{rG} = 69$ lb;
 $W_{rP} = W_{xG} = 207$ lb; $A_v = 9$
18. $D_G = 4.000$ in; $C = 2.625$ in; $VR = 40$
 $W_{xW} = W_{tG} = 462$ lb; $W_{xG} = W_{tW} = 53$ lb;
 $W_{rG} = W_{rW} = 120$ lb
 Efficiency = 70.3%; Worm speed = 1200 rpm;
 $P_i = 0.626$ hp
 $\sigma_G = 24\,223$ psi [Slightly high for phosphor bronze]
 Rated wear load = $W_{tr} = 659$ lb. [OK, $> W_{tG}$]

Chapter 11 Keys, Couplings, and Seals

1. Use 1/2 in square key; SAE 1040 cold-drawn steel; length = 3.75 in.
3. Use 3/8 in square key; SAE 1018 cold-drawn steel; required length = 1.02 in based on compression on cast iron hub; use $L = 1.50$ in to be just shorter than the 1.75 in hub length.
5. T = torque; D = shaft diameter; L = hub length. From Table 11–6, $K = T/(D^2L)$.
 (a) Data from Problem 1: required $K = 1313$; too high for any spline in Table 11–6.
 (c) Data from Problem 3: required $K = 208$; use 6 splines.

7. Sprocket: 1/2 in square key; SAE 1020 CD;
 $L = 1.00$ in
 Wormgear: 3/8 in square key; SAE 1020 CD;
 $L = 1.75$ in

13. $T = 2885$ lb·in

15. $T = 27\,970$ lb·in

19. Data from Problem 16: $T = 313$ lb·in per inch of hub length

Data from Problem 18: $T = 4300$ lb·in per inch of hub length

Chapter 12 Shaft Design

1. $T_B = 3436$ lb·in
 $F_{Bx} = W_{tB} = 430$ lb ←
 $F_{By} = W_{rB} = 156$ lb ↓
3. $T_B = 656$ lb·in
 $F_{Bx} = W_{tB} = 437$ lb →
 $F_{By} = W_{rB} = 159$ lb ↑
5. $T_D = 3938$ lb·in
 $W_{tD} = 985$ lb Up at 30° left of vertical
 $W_{rD} = 358$ lb To right 30° above horizontal
 $F_{Dx} = 182$ lb ←
 $F_{Dy} = 1032$ lb ↑
7. $T_C = 6563$ lb·in
 $F_{Cx} = W_{tC} = 1313$ lb →
 $F_{Cy} = W_{rC} = 478$ lb ↑
9. $T_C = 1432$ lb·in
 $F_{Cx} = W_{tC} = 477$ lb ←
 $F_{Cy} = W_{rC} = 174$ lb ↓
11. $T_F = 1432$ lb·in
 $W_{tF} = 477$ lb Down at 45° left of vertical
 $W_{rF} = 174$ lb To right 45° below horizontal
 $F_{Fx} = 214$ lb ←
 $F_{Fy} = 460$ lb ↓
13. $T_A = 3150$ lb·in
 $F_{Ax} = 0$
 $F_{Ay} = F_A = 630$ lb ↓
15. $T_C = 1444$ lb·in
 $F_C = 289$ lb Down at 15° left of vertical
 $F_{Cx} = 75$ lb ←
 $F_{Cy} = 279$ lb ↓
17. $T_C = 2056$ lb·in
 $F_{Cx} = F_C = 617$ lb ←
 $F_{Cy} = 0$
19. $T_C = 10\,500$ lb·in
 $F_{Cx} = F_C = F_{Dx} = F_D = 1500$ lb ←
 $F_{Cy} = F_{Dy} = 0$

21. $T_E = 1313 \text{ lb} \cdot \text{in}$
 $F_E = 438 \text{ lb}$ Up at 30° above horizontal
 $F_{Ex} = 379 \text{ lb} \rightarrow$
 $F_{Ey} = 219 \text{ lb} \uparrow$
31. $T_B = 727 \text{ lb} \cdot \text{in}$
 $F_{Bx} = W_{tB} = 351 \text{ lb} \rightarrow$
 $F_{By} = W_{rB} = 132 \text{ lb} \downarrow$
 $W_{xB} = 94 \text{ lb}$ exerts a CCW concentrated moment of $194.6 \text{ lb} \cdot \text{in}$ on shaft at B.
 W_{xB} also places the shaft in compression from A to B if bearing A resists the thrust load.
33. $T_A = 270 \text{ lb} \cdot \text{in}$
 $F_{Ax} = 0$
 $F_{Ay} = F_A = 162 \text{ lb} \downarrow$
 $F_{Cx} = W_{tW} = 265 \text{ lb} \leftarrow$
 $F_{Cy} = W_{rW} = 352 \text{ lb} \uparrow$
 $W_{xW} = 962 \text{ lb}$ exerts a CW concentrated moment of $962 \text{ lb} \cdot \text{in}$ on shaft at worm.
 W_{xW} also places the shaft in compression from bearing B to worm if bearing B resists the thrust load.
35. Torques: Shaft 1— $T_1 = 175 \text{ lb} \cdot \text{in}$; Shaft 2— $T_2 = 350 \text{ lb} \cdot \text{in}$; Shaft 3— $T_3 = 1050 \text{ lb} \cdot \text{in}$
 Forces for Shaft 2: Directions from end view for tangential; Side view of shaft for axial and radial
 Forces on Gear B: Tangential — $W_{tB} = 233 \text{ lb} \rightarrow$;
 Axial — $W_{xB} = 233 \text{ lb} \rightarrow$; Radial — $W_{rB} = 85 \text{ lb} \downarrow$
 Forces on Gear C: Tangential — $W_{tC} = 350 \text{ lb} \leftarrow$;
 Axial — $W_{xC} = 350 \text{ lb} \leftarrow$; Radial — $W_{rC} = 128 \text{ lb} \downarrow$
37. Torque on bevel gear: $T_g = 420 \text{ lb} \cdot \text{in}$;
 Torque on each sprocket: $T_{sp} = 210 \text{ lb} \cdot \text{in}$
 Bevel gear forces: Tangential — $W_{tg} = 108 \text{ lb}$;
 Radial — $W_{rg} = 12 \text{ lb}$; Axial — $W_{ag} = 37 \text{ lb}$
 Force on each sprocket:
 $F_{c2} = 42 \text{ lb} \rightarrow$; $F_{c1} = 42 \text{ lb} \leftarrow$
39. Torque on motor shaft: $T_m = 657 \text{ lb} \cdot \text{in}$;
 Bending force on motor shaft: $F_B = 352 \text{ lb}$
 Reducer input shaft 1: $T_1 = 986 \text{ lb} \cdot \text{in}$;
 Bending force from belt drive: $F_B = 352 \text{ lb}$
 Forces on Gear A: Tangential — $W_{tA} = 1096 \text{ lb}$;
 Radial — $W_{rA} = 399 \text{ lb}$
 Reducer shaft 2: $T_2 = 2958 \text{ lb} \cdot \text{in}$
 Forces on Gear B: Tangential — $W_{tB} = 1096 \text{ lb}$;
 Radial — $W_{rB} = 399 \text{ lb}$
 Forces on Gear C: Tangential — $W_{tC} = 1479 \text{ lb}$;
 Radial — $W_{rC} = 538 \text{ lb}$
 Reducer shaft 3: $T_3 = 5917 \text{ lb} \cdot \text{in}$
 Forces on output Gear D: Tangential — $W_{tD} = 1479 \text{ lb}$;
 Radial — $W_{rD} = 538 \text{ lb}$
 Forces on Chain sprocket: Net force
 $F_N = 2817 \text{ lb}$; $F_{Nx} = 964 \text{ lb}$; $F_{Ny} = 2647 \text{ lb}$

41. Force on Lever 2: $F_2 = 30 \text{ N}$;
 Torque = $1200 \text{ N} \cdot \text{mm}$ between B and D.

Chapter 13 Tolerances and Fits

- RC8: Hole—3.5050/3.5000; shaft—3.4930/3.4895; clearance—0.0070 to 0.0155 in
- RC8: Hole—0.6313/0.6285; shaft—0.6250/0.6234; clearance—0.0035 to 0.0079 in
- RC8: Hole—1.2540/1.2500; pin—1.2450/1.2425; clearance—0.0050 to 0.0115 in
- RC5: Hole—0.7512/0.7500; pin—0.7484/0.7476; clearance—0.0016 to 0.0036 in (tighter fit could be used)
- FN5: Hole—3.2522/3.2500; shaft—3.2584/3.2570; interference—0.0048 to 0.0084 in; pressure = 13 175 psi; stress = 64 363 psi
- FN5: Interference—0.0042 to 0.0072 in; pressure = 8894 psi; stress = 18 901 psi at inner surface of aluminum cylinder; stress = -8894 psi at outer surface of steel rod; stress in aluminum is very high
- Maximum interference = 0.001 78 in.
- Temperature = 567°F
- Shrinkage = 0.0038 in; $t = 284^\circ\text{F}$
- Final ID = 3.4973 in

Chapter 14 Rolling Contact Bearings

- Life = 2.76×10^6 rev
- $C = 12\,745 \text{ lb}$

For Problems 5–17 calling for the selection of suitable bearings for given applications, several possible solutions exist. The listed possible solutions use the data in Table 14–3 and the bearing with the smallest bore was selected that would satisfy the load requirements. The design life is a design decision and the values used for the given solutions are listed. For pure radial loads, R , the method from Section 14–9 is used. For both radial and thrust loading (R and T), the method from Section 14–10 is used. Because the data in Table 14–3 are not from any specific bearing manufacturer's catalog, these results cannot be relied upon for actual implementation. It is recommended that online manufacturer's catalogs be used for actual applications.

- For 10 000 h life:
 At B: $R = 4643 \text{ lb}$; Required $C = 33\,029 \text{ lb}$;
 Bearing 6319
 At C: $R = 2078 \text{ lb}$; Required $C = 14\,782 \text{ lb}$;
 Bearing 6311
- For 20 000 h life:
 At A: $R = 509 \text{ lb}$; Required $C = 2519 \text{ lb}$;
 Bearing 6302
 At C: $R = 1742 \text{ lb}$; Required $C = 8621 \text{ lb}$;
 Bearing 6212

9. For 20 000 h life: $R = 455$ lb;
Required $C = 5066$ lb; Bearing 6306
11. For 5000 h life: $R = 1265$ lb, $T = 645$ lb;
Required $C = 6284$ lb; Bearing 6307
13. For 15 000 h life: $R = 2875$ lb, $T = 1350$ lb;
Required $C = 31\,909$ lb; Bearing 6318
15. For 2000 h life: $R = 5.6$ kN, $T = 2.8$ kN;
Required $C = 25.61$ kN; Bearing 6306
17. For 20 000 h life: $R = 1.2$ kN, $T = 0.85$ kN;
Required $C = 20.62$ kN; Bearing 6305
19. Life = 45 285 hours
21. Life = 24 909 hours
23. Life = 39 231 hours
25. $C = 17\,229$ lb
27. $C = 4580$ lb

Chapter 15 No practice problems

Chapter 16 Plain Surface Bearings

All problems in this chapter are design problems for which no unique solutions exist.

Example Solution—Problem 16–1:

Bearing length = $L = 1.50$ in;
Bearing bore diameter = $D = 3.00$ in
Pressure = $p = 16.67$ psi; $V = 1374$ ft/min
 $pV = 22\,900$ psi-fpm;
Design value for $pV = 45\,800$ psi-fpm
Specify porous bronze/oil impregnated bearing:
 pV rating = 50 000 psi-fpm

Chapter 17 Linear Motion Elements

5. $2\frac{1}{2}$ –3 Acme thread
6. $L > 1.23$ in
7. $T = 6974$ lb·in
8. $T = 3712$ lb·in
11. Lead angle = 4.72° ; self-locking
12. Efficiency = 35%
13. $n = 180$ rpm; $P = 0.866$ hp
14. Specify a $\frac{3}{4}$ –2 ball screw
17. 24.7 years
18. Required $A_t = 1667$ mm²; Use M55 $\times$ 9 screw
19. $T_u = 658.6$ N·m
20. $T_d = 291.9$ N·m
21. Power = 260.5 kW
22. Lead angle = $\lambda = 3.25^\circ < 5.0^\circ$ – Self locking
23. Efficiency = 33.4 percent

Chapter 18 Springs

1. $k = 13.3$ lb/in
2. $L_f = 1.497$ in
3. $F_s = 47.8$ lb; $L_f = 1.25$ in
7. $ID = 0.93$ in; $D_m = 1.015$ in; $C = 11.94$;
 $N = 6.6$ coils
8. $C = 8.49$; $p = 0.241$ in; pitch angle = 8.70° ;
 $L_s = 1.12$ in
9. $F_o = 10.25$ lb; stress = 74 500 psi
11. $OD = 0.583$ in when at solid length
12. $F_s = 26.05$ lb; stress = 189 300 psi (High)
31. Bending stress = 114 000 psi;
torsion stress = 62 600 psi. Stresses are safe.
35. Torque = 0.91 lb·in to rotate spring 180° .
Stress = 184 800 psi; OK for severe service

Chapter 19 Fasteners

4. Grade 2 bolts: 5/16–18; $T = 70.3$ lb·in
5. $F = 1190$ lb
6. $F = 4.23$ kN
7. Nearest metric thread is M24 $\times$ 2. Metric thread is 1.8 mm larger (8% larger).
8. Closest standard thread is M5 $\times$ 0.8. (#10–32 is also close.)
9. $F = 6.35$ kN
10. (a) 1177 lb
(c) 2385 lb
(e) 2862 lb
(g) 1081 lb
(i) 2067 lb
(k) 143 lb

Chapter 20 Machine Frames, Bolted Connections, and Welded Joints

Problems 1–16 are design problems for which there are no unique solutions.

17.

	Material	Diameter (in)	Weight (lb per inch of length)
a.	1020 HR steel	0.638	0.0906
c.	Aluminum 2014–T6	0.451	0.0160
e.	Ti–6Al–4V (Annealed)	0.319	0.0128

Chapter 21 Electric Motors and Controls

13. 480V, 3phase because the current would be lower and the motor size smaller
16. $n_s = 1800$ rpm in the United States
 $n_s = 1500$ rpm in France
17. 2-Pole motor; $n = 3600$ rpm at zero load (approximate)
18. $n_s = 12\,000$ rpm
19. 1725 rpm and 1140 rpm
20. Variable frequency control
34. (a) Single phase, split phase AC motor
(b) $T = 41.4$ lb·in
(c) $T = 62.2$ lb·in
(d) $T = 145$ lb·in
35. (b) $T = 4.15$ N·m
(c) $T = 6.23$ N·m
(d) $T = 14.5$ N·m
39. Full load speed = synchronous speed = 720 rpm
47. Use a NEMA Type K SCR control to convert 115 VAC to 90 VDC; use a 90 VDC motor.

51. Speed theoretically increases to infinity
52. $T = 20.5$ N·m
54. NEMA 2 starter
55. NEMA 1 starter

Chapter 22 Motion Control: Clutches and Brakes

1. $T = 495$ lb·in
3. $T = 41$ lb·in
5. Data from Problem 1 : $T = 180$ lb·in
Data from Problem 3 : $T = 27.4$ lb·in
7. Clutch: $T = 2122$ N·m
Brake: $T = 531$ N·m
8. $T = 143$ lb·ft
9. $T = 60.9$ lb·ft
11. $T = 223.6$ lb·ft
15. $F_a = 109$ lb
17. $W = 138$ lb
18. $b > 16.0$ in

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